

Juan David Muñoz-Bolaños

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Iventim Solutions GmbH

Hiring Manager

München

Deutschland

Subject: Application for Optical Metrology Engineer

Dear Hiring Manager,

As a physicist and optical engineer with a focus on precision metrology, I am applying for the position of Optical Metrology Engineer in your team. My focus is on leading measurement systems from concept to automated, functional prototypes – a strength I can optimally contribute to your R&D projects.

During my PhD, I designed an automated wavefront measurement system from scratch and controlled it with Python/C++. Daily modeling and tolerance analysis with ZEMAX OpticStudio were as much part of my repertoire as building a Raman spectrometer at IPHT Jena. This end-to-end approach ensures precise measurement results and well-founded design decisions.

Fully integrated into the DACH work culture, I will be available at short notice following the completion of my PhD in May 2026. I look forward to bringing my experience in optical automation to Iventim.

Sincerely,

Juan David Muñoz-Bolaños

Optical Metrology Engineer

Medizinische Universität Innsbruck

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Optical Metrology Engineer

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Professional Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

TECHNICAL COMPETENCIES

Optical Metrology & Simulation

Expert in wavefront sensing, phase retrieval, and ZEMAX OpticStudio tolerancing. Proficient in characterizing aberration sensitivity of high-NA microscope objectives.

System Automation & Control

Experience building C++/Python/LabVIEW control stacks for adaptive optics, scientific cameras (sCMOS), and opto-mechanical stages.

Data Science for Optics

Application of supervised ML and AI algorithms to hyperspectral imaging data. Proficient with NumPy, Pandas, Scikit-learn, and JAX for high-performance computation.

Hardware Integration & Sourcing

End-to-end R&D experience in multi-photon microscope assembly. Competent in technical component specification (Thorlabs, Edmund Optics) and incoming QA.

Process Reliability

Hands-on experience with industrial vision solutions (Halcon) and ensuring measurement traceability via Gage R&R assessments.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

PhD Researcher - Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **Automated Metrology Setups:** Designed and built an automated wavefront sensing system; developed the full software control stack to unify scientific cameras and wavefront modulators.
- **Optical Modeling & Design:** Performed systematic ZEMAX tolerance analysis to optimize the process of custom objective lens characterization and aberrometry.
- **Integration & Alignment:** Achieved diffraction-limited performance in a custom-built nonlinear microscope by precisely aligning galvo scanners, femtosecond lasers, and PMT detectors.
- **Technical Documentation:** Managed technical specification sheets and qualified vendors (Hamamatsu, Thorlabs) for high-precision opto-mechanical integration.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher (Photonics & Data)

Jena, Germany

Jun 2020 – Dec 2021

- **Spectrometer Build:** Handled the end-to-end optical layout and detector calibration of a 1064 nm fiber-probe Raman spectrometer prototype.
- **AI-Driven Data Analysis:** Leveraged custom Python packages for Machine Learning classification on hyperspectral tissue images, accelerating experimental turnaround.
- **Published Research:** Contributed to the development of non-invasive clinical diagnostic tools, results documented in peer-reviewed journals (Applied Sciences, ACS Photonics).

INTECOL S.A.S.
Junior Machine Vision System Engineer

Medellín, Colombia
Mar 2018 – Mar 2019

- **Industrial Machine Vision:** Engineered Halcon inspection systems for beverage production lines; solved critical label and sealing quality issues via custom illumination design.
- **Hardware Commissioning:** Optimized robotic vision for auto-part recognition using smart cameras and tailored illumination methods, directly impacting production efficiency at a car subdivision of Renault.
- **Gage R&R Assessments:** Verified system accuracy and process stability within industrial quality control frameworks.

EDUCATION

Medical University of Innsbruck <i>Ph.D. in Adaptive Optics & Microscopy</i>	Austria Jan 2022 – Present
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Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany 2019 – 2021
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University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia 2012 – 2018
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PUBLICATIONS & AWARDS

- **Awards:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
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Group leader: Spectroscopy Imaging
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